

Benchmark #3

3MeV Xe on UO2

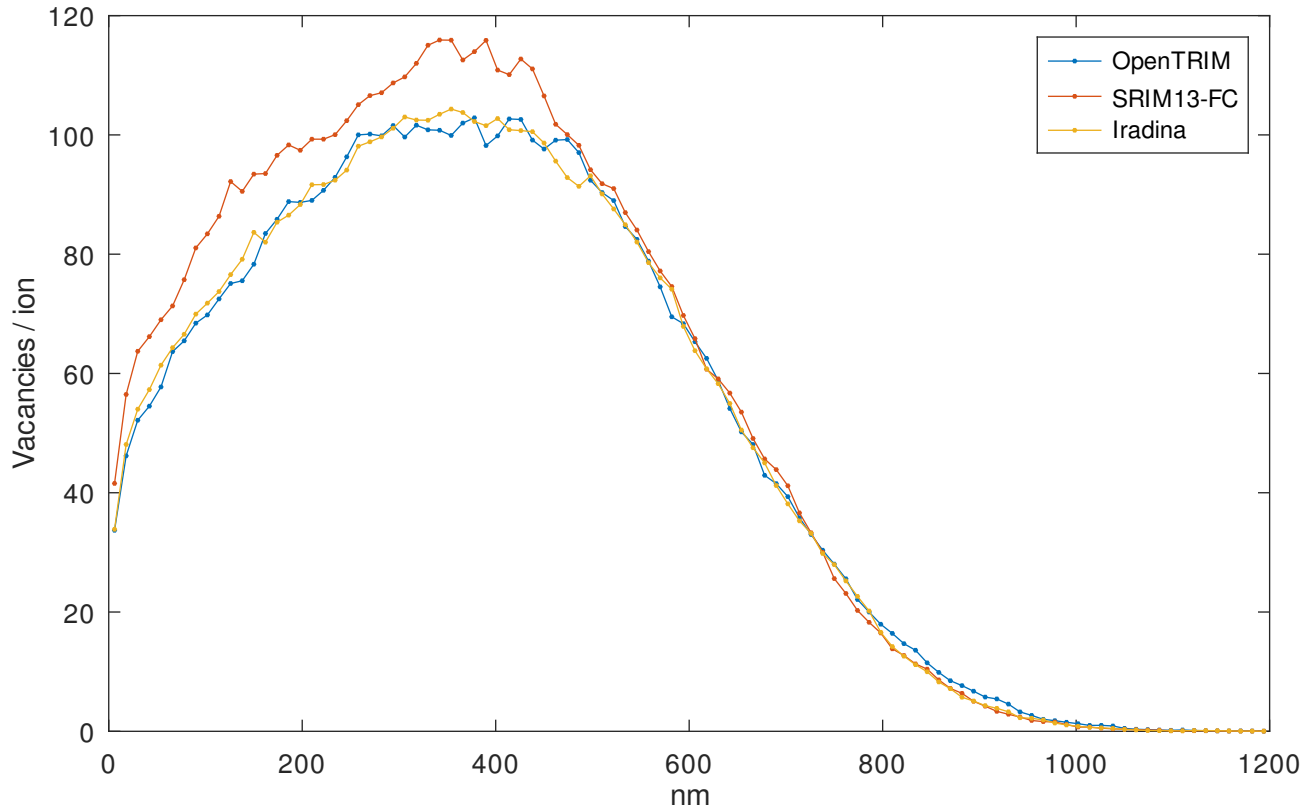
Ion energy E0 = 3e+06 eV

Target depth = 1200 nm

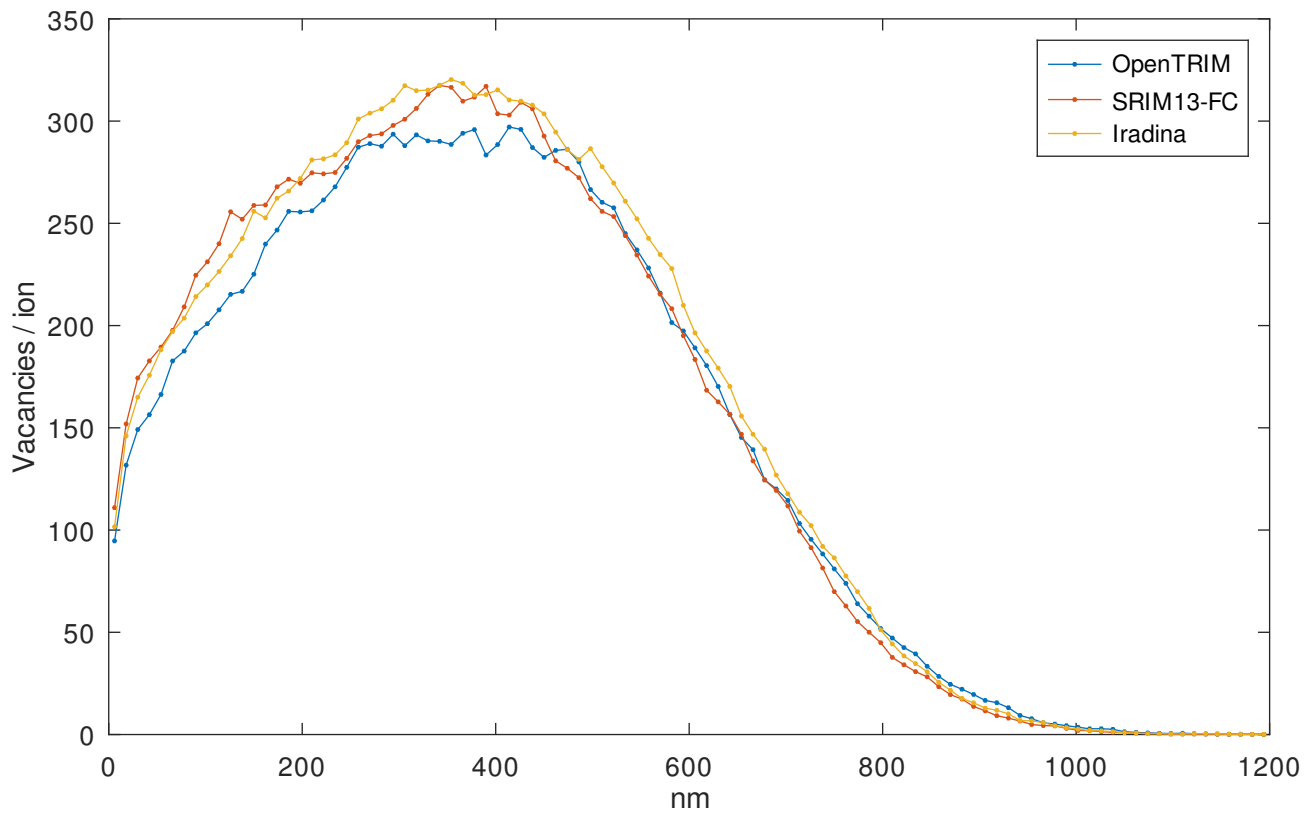
Summary Table

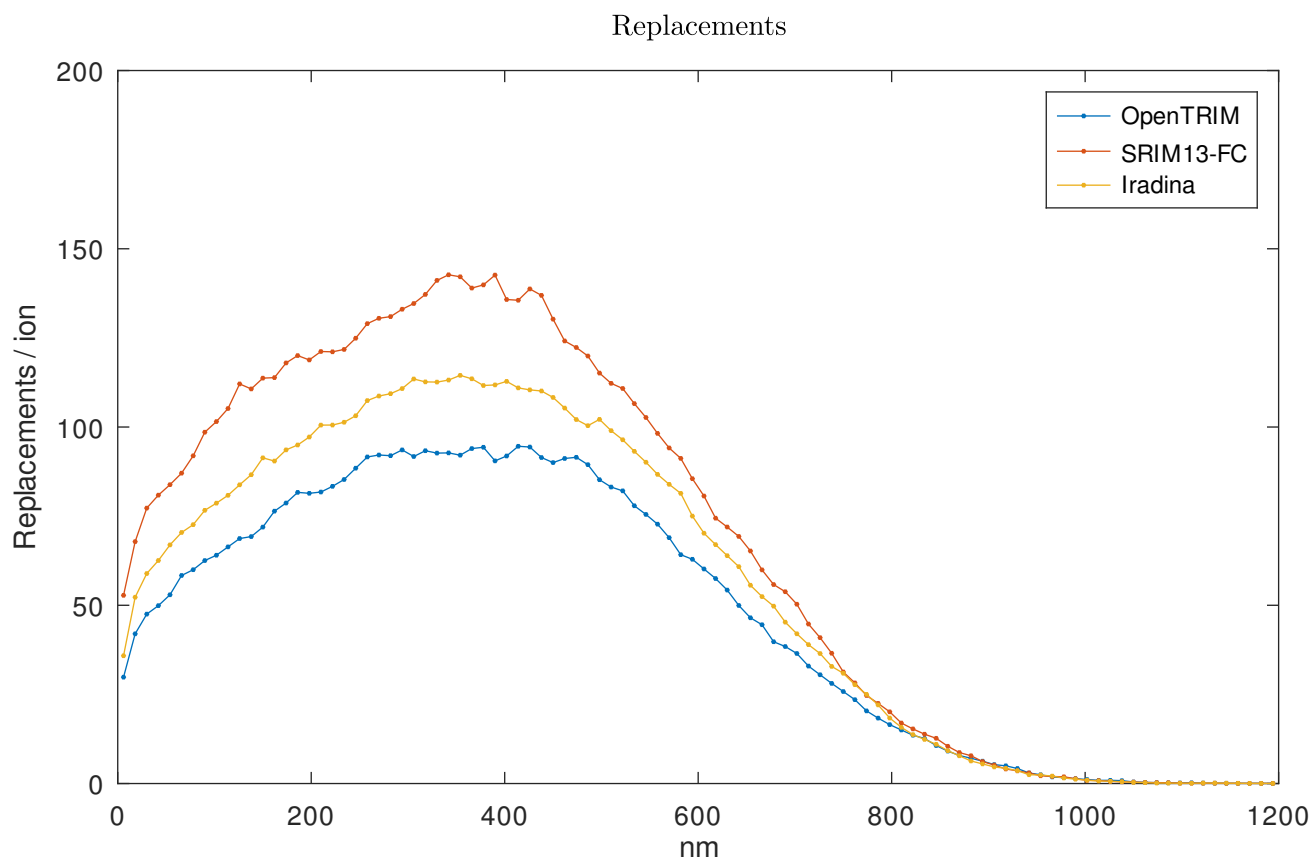
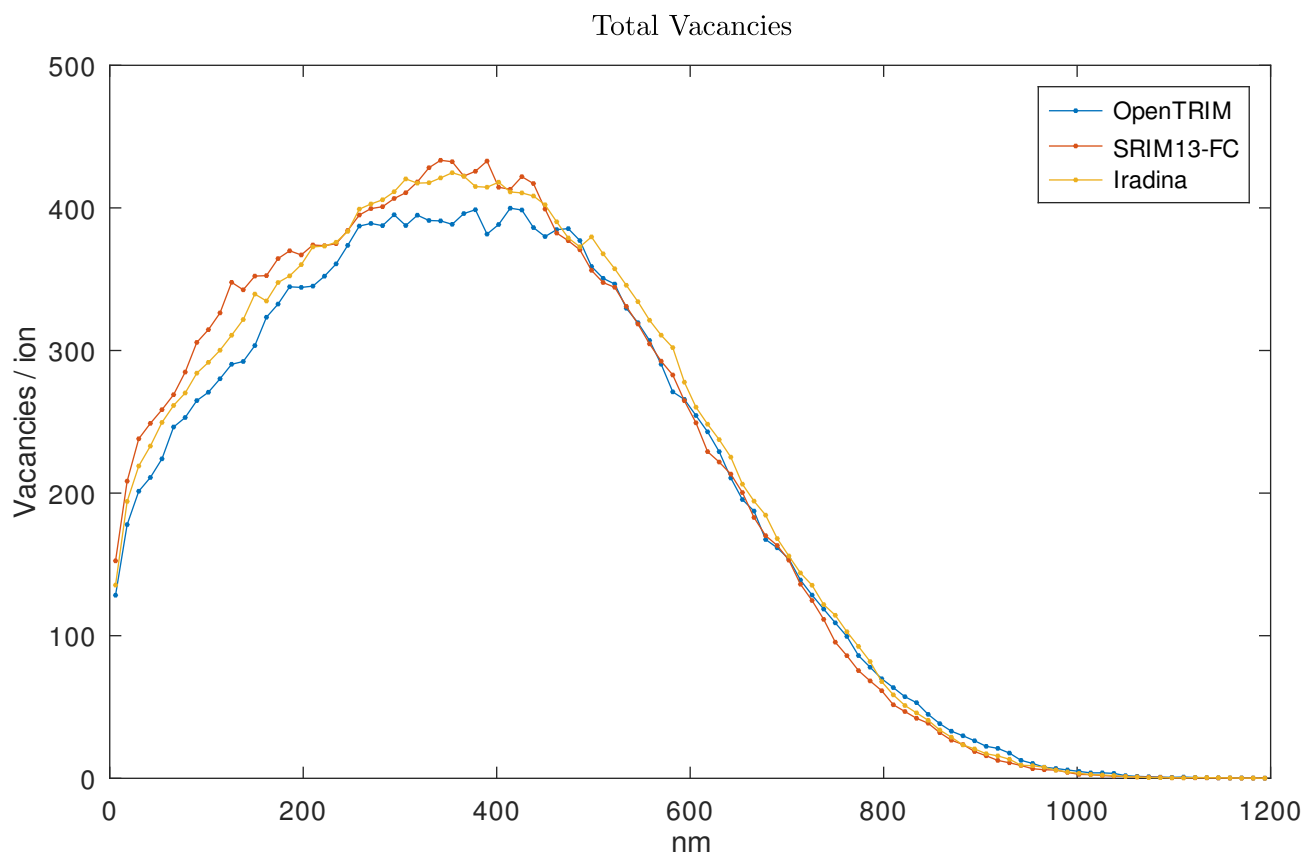
Quantity	OpenTRIM	SRIM13-FC	Iradina
V(U)	5.06e+03	5.48e+03	5.06e+03
V(O)	1.46e+04	1.51e+04	1.56e+04
V(tot)	1.97e+04	2.06e+04	2.06e+04
R(tot)	4.65e+03	6.7e+03	5.55e+03
I(Xe)	0.996	0.995	0.995
$\Delta E_e(\text{Xe})/E_0$	0.3	0.294	0
$\Delta E_e(r)/E_0$	0.32	0.318	0
$\Delta E_e(\text{tot})/E_0$	0.62	0.613	0.634
$\Delta E_{ph}(\text{Xe})/E_0$	0.00196	0.00175	0
$\Delta E_{ph}(r)/E_0$	0.375	0.36	0
$\Delta E_{ph}(\text{tot})/E_0$	0.377	0.362	0.363
$1 - (\Delta E_e + \Delta E_{ph})/E_0$	0.00237	0.0259	0.00242

Vacancies of U in Uranium oxide

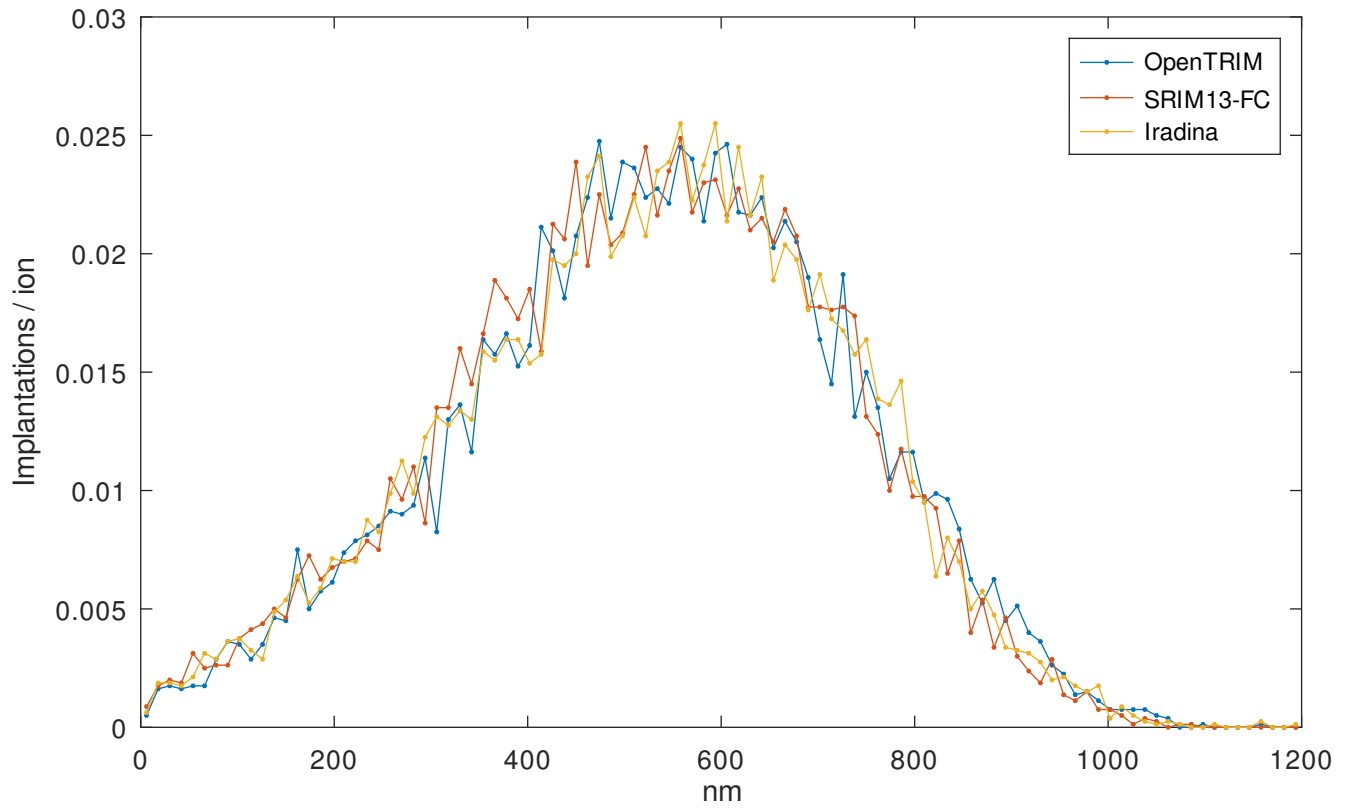


Vacancies of O in Uranium oxide

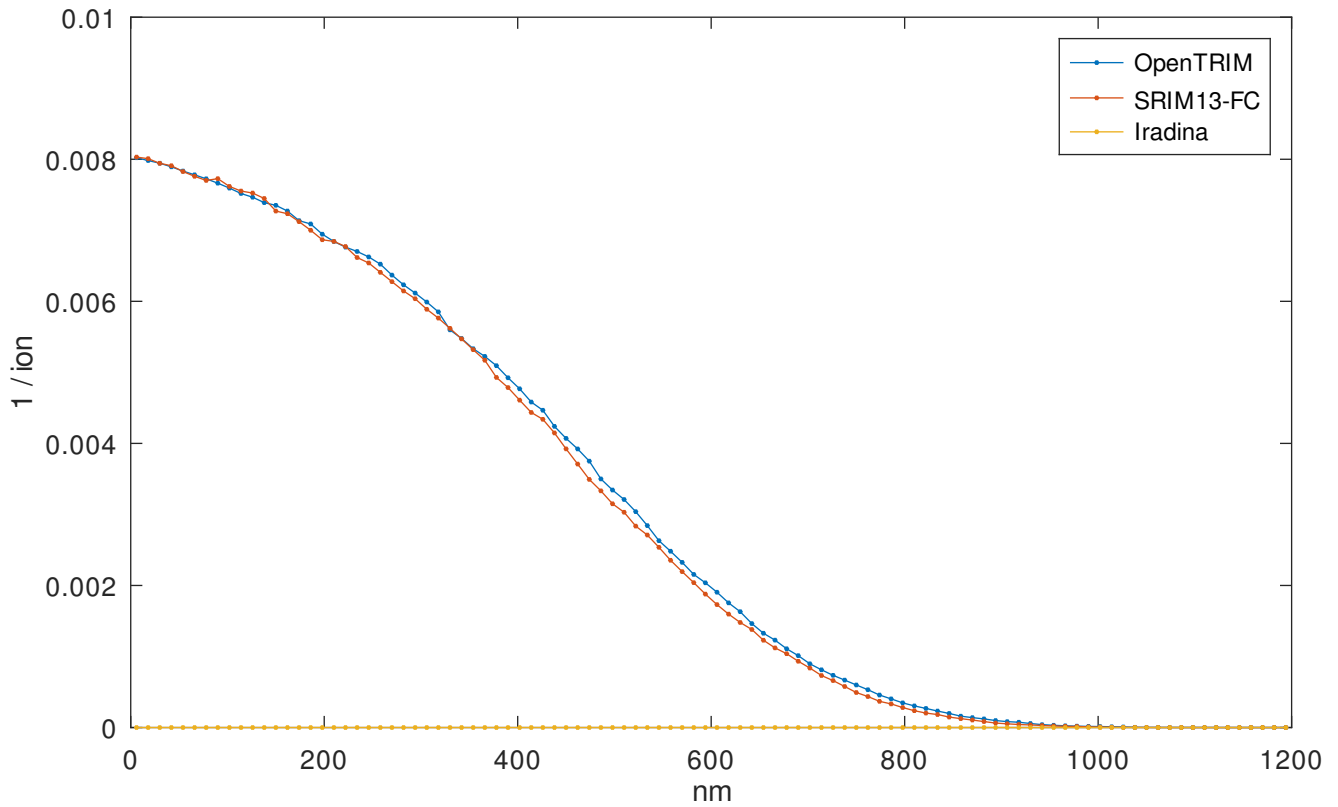




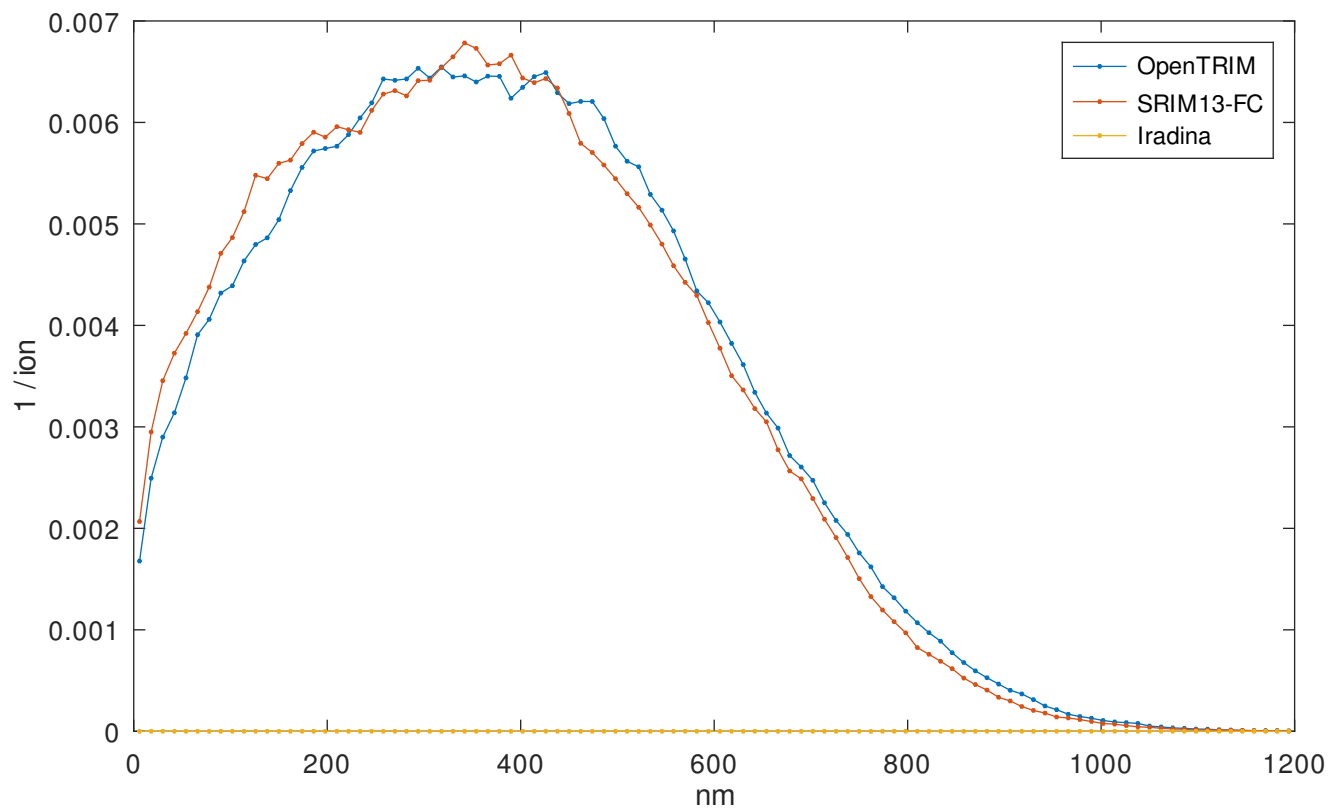
Implanted Xe ion



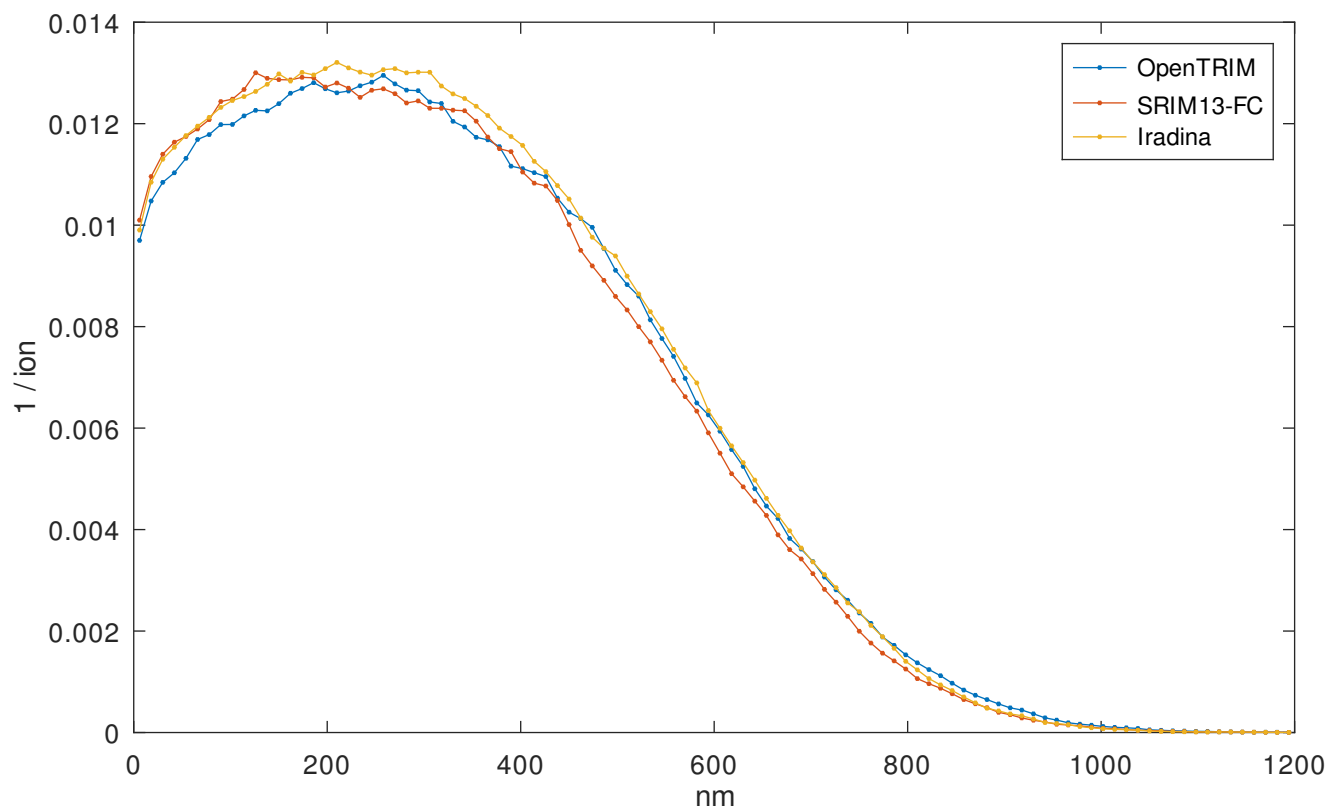
Electronic energy loss fraction $\Delta E_e/E_0$ by Xe ion



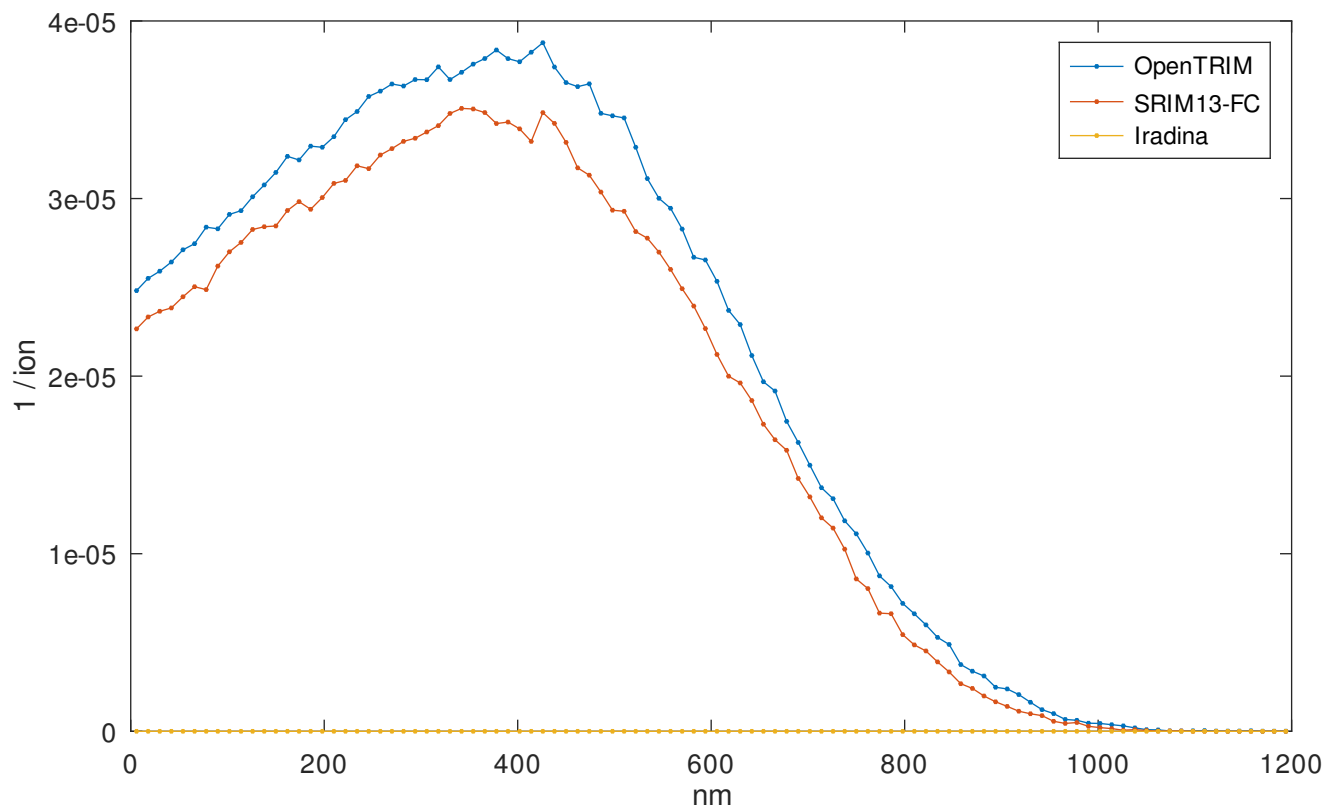
Electronic energy loss fraction $\Delta E_e/E_0$ by recoils



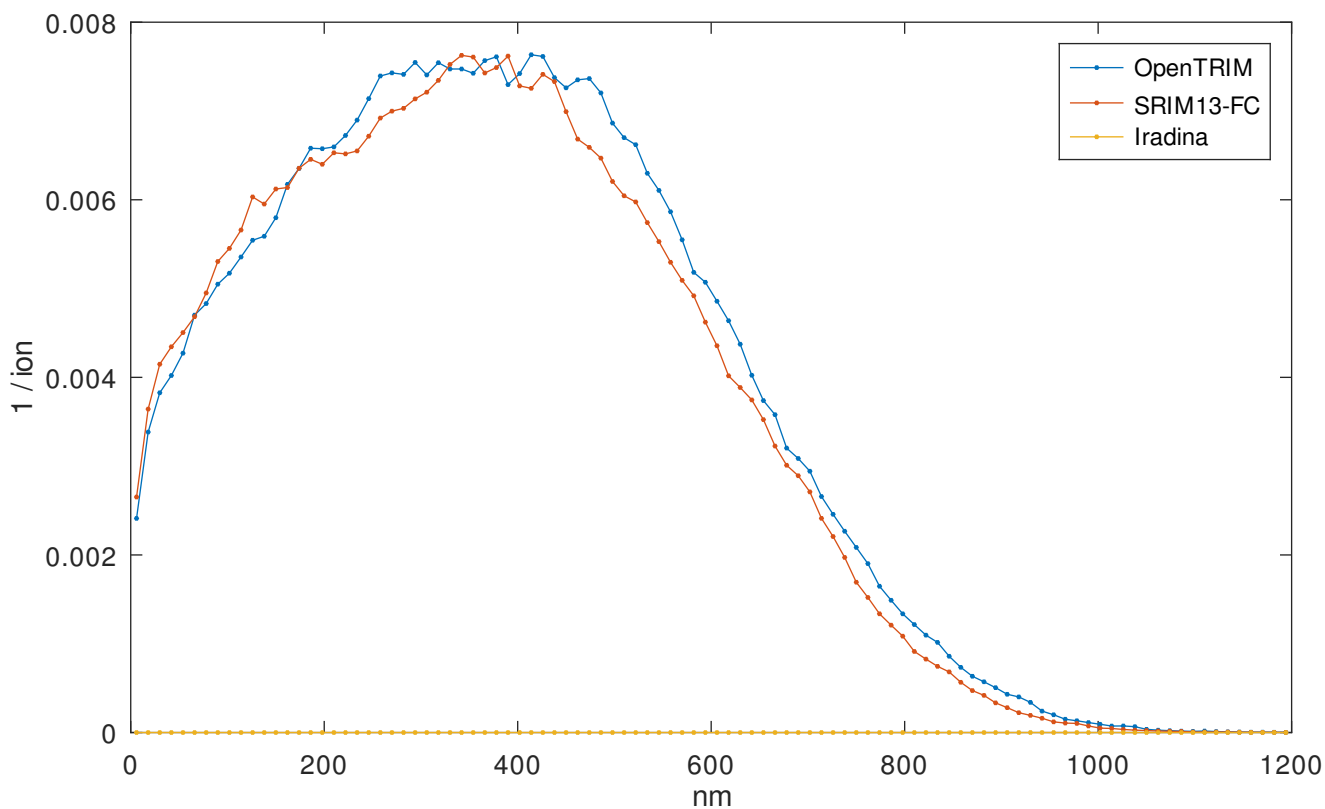
Total Electronic energy loss fraction $\Delta E_e/E_0$



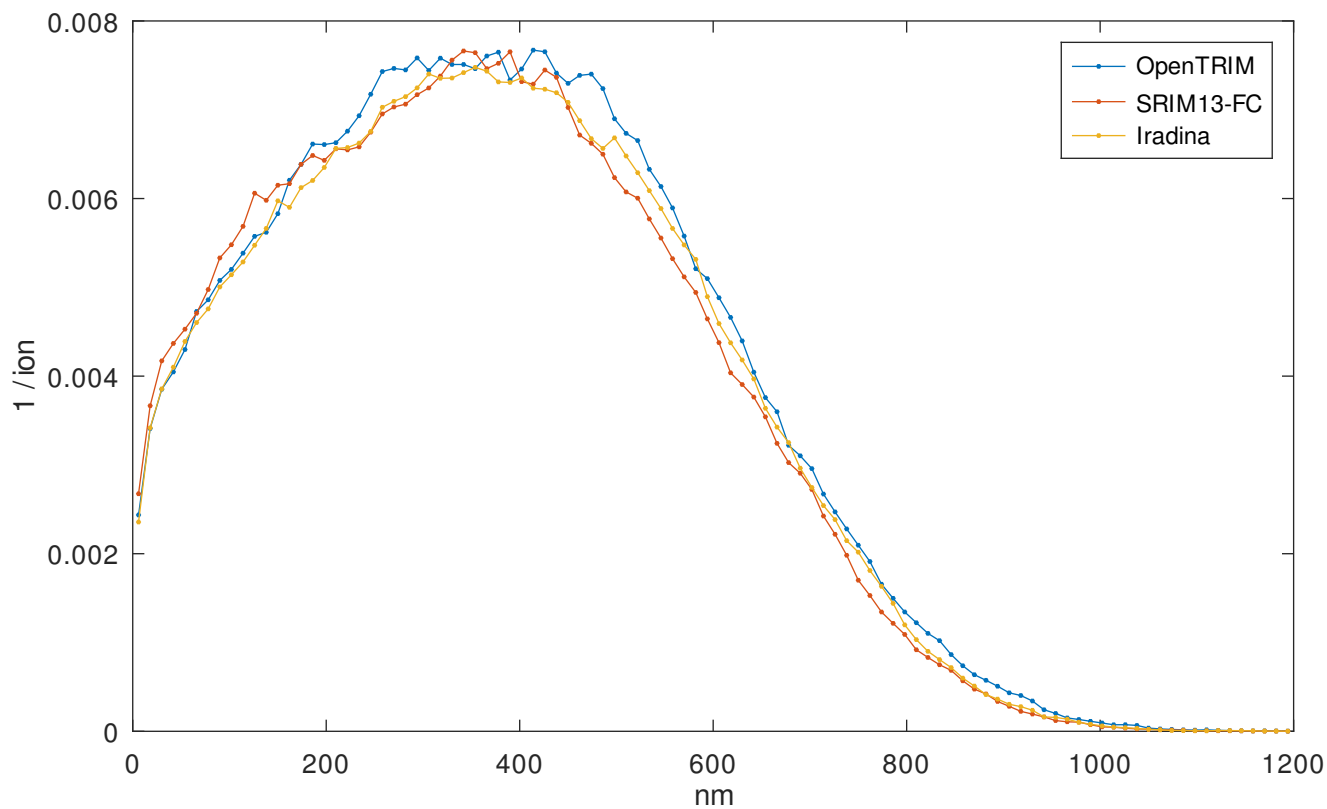
Phonon energy fraction $\Delta E_{Ph}/E_0$ by Xe ion



Phonon energy fraction $\Delta E_{Ph}/E_0$ by recoils



Total Phonon energy fraction $\Delta E_{Ph}/E_0$



Total fractional energy deposition $(\Delta E_e + \Delta E_{Ph})/E_0$

